

TrES-2b

R-1169

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_190506 · created 2026-07-31T03:09:55+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458609.79939 +/- 0.00065 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.149 +/- 0.011
Transit depth (Rp/Rs)^2	2.21 +/- 0.32 [%]
Orbital Inclination (inc)	83.94 +/- 0.53
Airmass coefficient 1 (a1)	1.007 +/- 0.012
Airmass coefficient 2 (a2)	-0.0031 +/- 0.0048
Scatter in the residuals of the lightcurve fit is	1.15 %
Best Comparison Star	#3 - [81, 116]
Optimal Aperture	4.52
Optimal Annulus	8.29
Transit Duration (day)	0.0778 +/- 0.0081

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.149 ± 0.011 vs 0.1254 (deviation 1.91σ)
Duration fit vs published	0.0778 d vs 0.0746 d (deviation 0.35σ)
Mid-time O–C	-1.6 min
Depth SNR	12.1
Residual scatter	1.15 %

Light curve

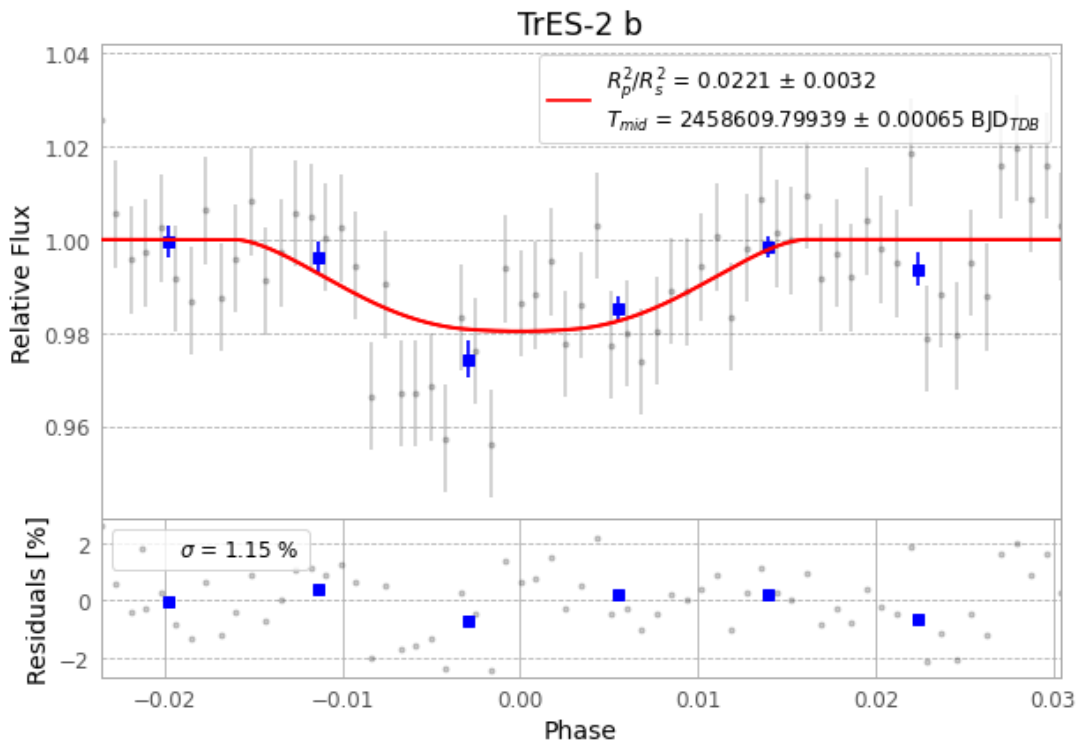


Figure 1 — FinalLightCurve_TrES-2 b_2019-05-05.png (SHA-256 a8abd75daf2441a2...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [386, 72], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 87628f9bd23021f7...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

66 FITS frames (43 MB), TRES-2190506054212.FITS → TRES-2190506085712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-190506.log (b3d400cab824...), FinalLightCurve_TrES-2 b_2019-05-05.png (a8abd75daf24...), FinalLightCurve_TrES-2 b_2019-05-05.csv (0ef9ff43ebf7...)

Compute resources

Wall time 140.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 03:09Z.